LOCALISATIONS OF AN ABSOLUTELY INTEGRALLY CLOSED RING ARE STRICTLY HENSELIAN

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Absolute integral closedness passes to quotients and localisations. Lemma \ref{lemma-absolutely-integrally-closed-quotient-localization}, cited at proof lines 3611-3612 and 3614-3615. Reduces to the local case and gives algebraic closedness of the residue field.
- Definition of a henselian ring via factorisation of monic polynomials. Algebra, Definition \ref{algebra-definition-henselian}, cited at proof lines 3616-3617. Is the condition verified in the reduced case.

Source reference. Original lemma 0DCS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16742–16742. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0DCN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16738–16738. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 04GF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5831–5831. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let A be absolutely integrally closed. Let $\mathfrak{p} \subset A$ be a prime. Then the local ring $A_{\mathfrak{p}}$ is strictly henselian.*

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Proof. By Lemma 0DCN we may assume A is a local ring and $\mathfrak{p}$ is its maximal ideal. The residue field is algebraically closed by Lemma 0DCN. Every monic polynomial decomposes completely into linear factors hence Algebra, Definition 04GF applies directly. □